
∞ MMLA: How MEMORY LETS THE PAST SHAPE THE FUTURE

PUBLIC MANUSCRIPT OF MMLA

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Project: github.com/MMLA-org/mmla-memory

ABSTRACT

Proposal. Long context can replay history, but it does not decide which completed observations deserve authority. MMLA formalizes a bounded resident memory between transient context and slow weight updates. A completed local segment is eventized; for each event, a target-conditioned constructor proposes semantic content and a trusted assembler produces a complete versioned row; deployment either commits that row atomically or returns NULL. Realized futures may price actions during training, while deployment remains causal and future-blind.

Validated components. Controlled studies establish narrower ingredients. Lifecycle execution is exact on 300/300 held-out records for each of three seeds. Calibrated selection with full-archive fallback improves over a weak budget-matched dense baseline by 5.5–16.6 F1 and over BM25 by 4.0–6.2 F1 on held-out multi-hop QA; the original Llama budget execution is retained as failed, while the corrected Qwen packer satisfies the stated per-record caps. Typed anchor–filler transport reaches 240/240 held-out exactness per seed while three same-checkpoint controls obtain 0/240 whole-record successes.

Current blocker. The integration loop is not complete. Dense-row, structured-span, and checkpoint-native readers trained from V28—the frozen 352.3M-parameter model-only checkpoint produced by a 50.0M-token native-scaffold pilot—all fail semantic qualification across three seeds. Candidate exactness is 0–45/23,040, query exactness is 10–1,536/9,216, and record-macro Brier remains near the 1,025-class uniform reference. Structural mapping passes, but none of the nine jobs qualifies. Predictive overwrite is therefore closed by gate.

MMLA: architecture, admitted evidence, and the closed loop

four layers are kept separate: formal architecture, implemented ancestors, validated components, and unverified successors

The diagram separates the proposed architecture from admitted component evidence and unresolved integration.

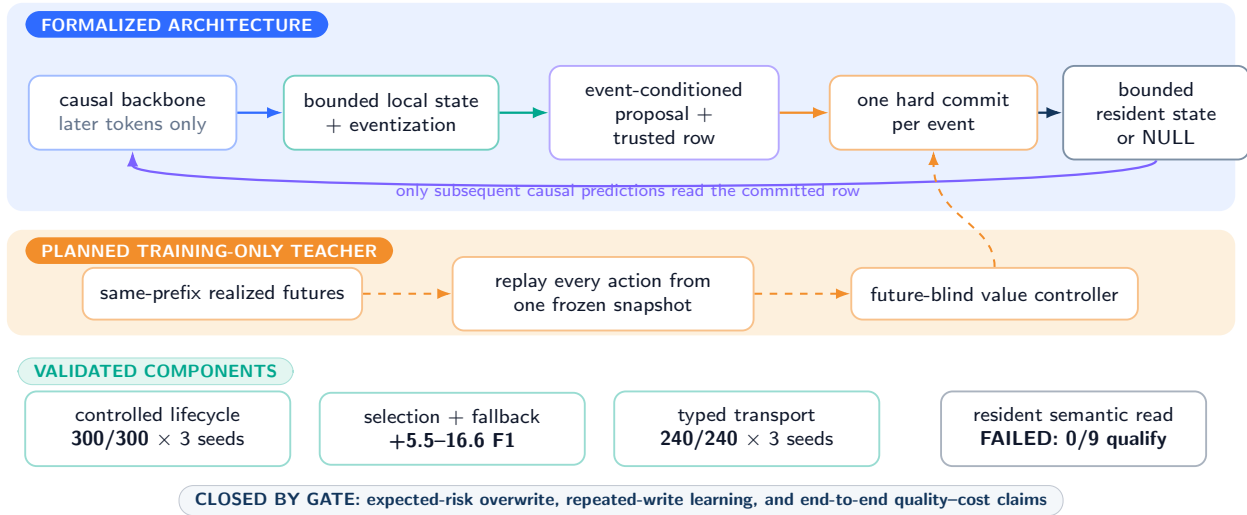


Figure 1: **Proposal, evidence, and present boundary in one view.** Solid arrows show the deployed causal path; the orange branch exists only during training. The bottom row reports admitted component evidence and the matched semantic-interface failure. The complete predictive-overwrite loop has not yet been validated.

Reading guide. This v3 replacement is a condensed, self-contained public presentation of the MMLA research program. It preserves the numerical evidence admitted under the project’s frozen preregistered audit protocols through 12 August 2026. Formalized architecture, implemented ancestors, validated components, registered failures, and unverified successors are labeled separately. The complete predictive-overwrite loop remains unvalidated. The public project page and companion Technical Report are available at <https://github.com/MMLA-org/mmla-memory>.

1 Why Memory Management Is a Different Problem

We retain the project name MMLA from the earlier *Memory-Managed Long-Context Attention* formulation. The present architecture is mixer-agnostic and is not restricted to attention alone.

Transformers and their long-context successors have made remarkable progress in retaining or accessing more tokens [32, 16, 12, 30, 24, 11, 25, 39, 38]. Yet an agent running for months faces a problem that context length alone does not solve. Its history may contain obsolete values, mutually inconsistent versions, malicious writes, or facts whose later importance was impossible to know when they first appeared. Reprocessing every token increases cost without deciding which information deserves authority.

The object of MMLA is therefore a bounded, revisable state between transient context and slow weight updates. It should support at least five behaviors: incorporate a useful new fact, overwrite a stale version, preserve unrelated state, delete or tombstone a record, and abstain when the evidence for writing is weak. A large retrieval archive may complement that state, but archive search and authoritative memory mutation are different operations.

The hardest decision is the write. Immediate attention or salience measures whether information matters *now*; memory should reflect whether it is likely to matter *later*. Training trajectories reveal actual futures that deployment lacks. This motivates a teacher that prices candidate actions from realized futures and a deployable controller that predicts those prices from completed history alone. Predictive information, predictive state, and information bottleneck ideas provide useful language for this objective [31, 4, 20, 1]. They do not prove that the objective is learnable.

We make three claims, at different levels:

1. **Architectural claim.** A causal language model, bounded completed-segment consolidation, one atomic overwrite or NULL per event, and training-only future supervision can be composed without deploy-time future leakage.
2. **Component evidence.** Controlled experiments support lifecycle execution, calibrated fallback, and typed relational transport under their stated boundaries.
3. **Integration boundary.** Three current-state interfaces fail semantic qualification on the same frozen release. This blocks predictive overwrite training and any end-to-end success claim.

Claim–evidence map: one vocabulary for every layer

status describes the strongest justified statement; it does not transfer upward to the full architecture

CLAIM	ADMITTED BASIS	STATUS
Causal completed-segment boundary	equations + interface contract; no deploy future	FORMALIZED
Controlled lifecycle executor	version, lock, tombstone, eviction implementation	IMPLEMENTED
Lifecycle and typed binding	300/300 and 240/240 per seed under controls	VALIDATED
Resident-row semantic read	PM-S1, PM-I1, PM-I2; PM-I2 has 0/9 qualified	FAILED
Expected-risk overwrite controller	semantic prerequisite failed; oracle never opened	CLOSED BY GATE
Dual-view authoritative row	recommended vNext contract motivated by negatives	PLANNED
Full-system quality–cost claim	no repeated-write or full-system crossover test	PLANNED

The three semantic-interface failures close predictive overwrite without implying that the entire MMLA research program has failed.

Figure 2: **Claim–evidence matrix.** Status is attached to the strongest justified statement, not inherited by the whole architecture. “Implemented” denotes a mechanical ancestor, while “validated” requires evidence admitted under the frozen protocol.

2 Predictive Memory Consolidation

2.1 Three state scales and a slower parameter scale

Let $x_1, x_2, \dots$ be a token stream. The model maintains a causal hidden state h_t , a completed local workspace B_n , and S persistent rows $M_n = [r_{n,1}; \dots; r_{n,S}]$ between segment boundaries. Model weights form a slower time scale. The architecture family does not require a permanent 1:3:9 ratio, 2,048 rows, or 50-turn delay. To make the first end-to-end test reproducible, however, we fix a proposed *MMLA-R0* reference: 12 mixer blocks in four repetitions of [local attention, recurrent/linear state, memory read], a 2,048-token local window, 256-token completed segments, $S = 256$ resident rows, at most four ordered events per segment, and at most one hard row commit per event. The archive is disabled during semantic-interface qualification. Training uses BF16 for neural computation with FP32 normalization and controller reductions. This is a falsification configuration, not an empirically optimized product.

A later budget-aware block may learn how to mix the three transient paths,

$$y_t = \sum_{j \in \{\text{local}, \text{linear}, \text{memory}\}} w_{t,j} y_{t,j}, \quad w_t = \text{softmax } r_\theta(h_t, M_{n(t)-1}, b_t), \quad (1)$$

where b_t describes the compute budget. The invariant is not a universal mixer ratio; it is that resident state is bounded, explicit, and editable.

Formal MMLA architecture: ordered events and one hard commit per event

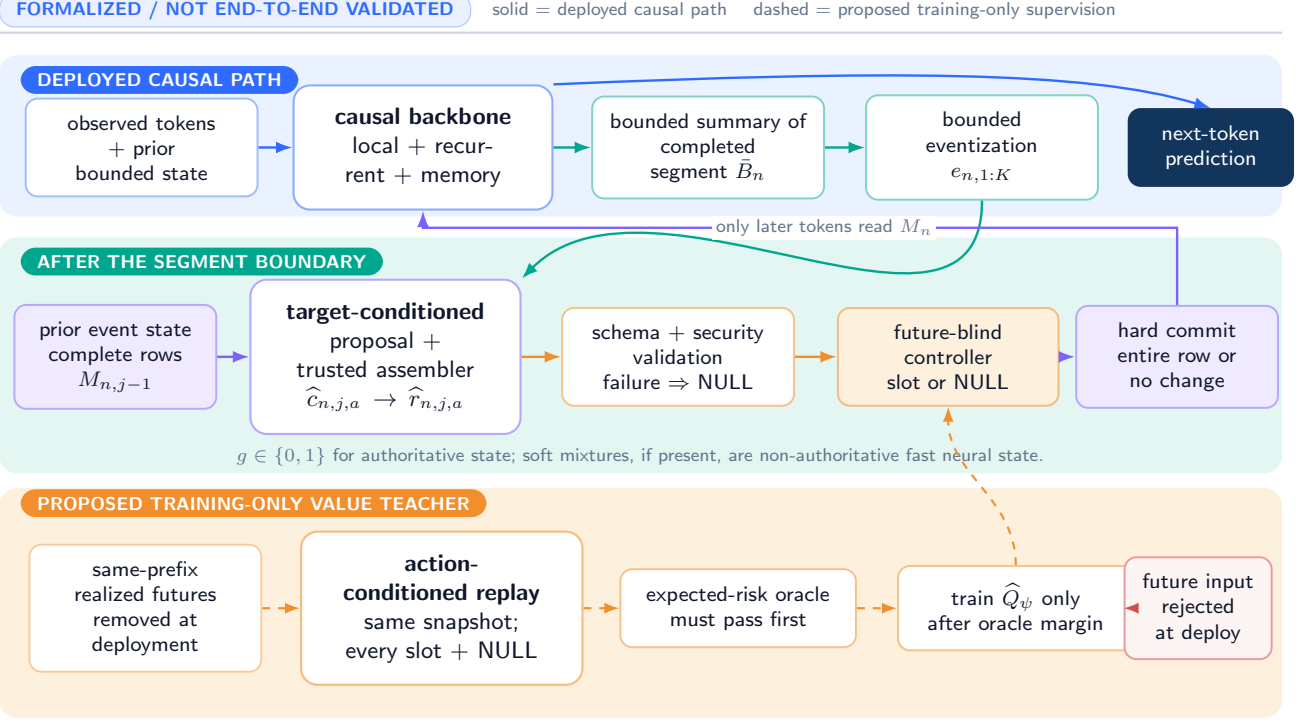


Figure 3: **Causal architecture and training-only credit path.** The model predicts from prior state. Bidirectional processing is restricted to an already completed segment, and its update can affect only later tokens. Same-prefix futures and action-conditioned replay supervise values during training; deployment receives none of those futures.

2.2 Completed-segment bidirectionality without leakage

Suppose segment n occupies $(b_{n-1}, b_n]$. Prediction remains autoregressive,

$$p_\theta(x_{1:T}) = \prod_{t=1}^T p_\theta(x_t \mid x_{<t}, M_{n(t)-1}). \quad (2)$$

Only after x_{b_n} is observed may a bounded consolidator form

$$\bar{B}_n = C_\phi(h_{b_{n-1}+1:b_n}, M_{n-1}), \quad M_n \rightarrow p_\theta(x_t \mid x_{<t}) \quad (t \leq b_n). \quad (3)$$

It may use bidirectional attention *inside that completed segment*; it cannot see $X_{>b_n}$ and cannot alter emitted logits. This is retrospective encoding, not bidirectional generation. Eventization then produces an ordered bounded list

$$E_\eta(\bar{B}_n) = (e_{n,1}, \dots, e_{n,K_n}), \quad K_n \leq K_{\max}. \quad (4)$$

A multi-fact segment is therefore decomposed before one-row actions are considered. A true multi-row atomic transaction is deferred rather than hidden inside one action.

Within the boundary, let $M_{n,0} = M_{n-1}$ and process events in order. The state after event j is $M_{n,j}$; the segment-level state exposed to later tokens is $M_n = M_{n,K_n}$.

2.3 A complete target-conditioned row

For event $e_{n,j}$ and target $a \in \{1, \dots, S\}$, a neural constructor proposes semantic content,

$$\hat{c}_{n,j,a} = G_\phi(\bar{B}_n, r_{n,j-1,a}, M_{n,j-1}, e_{n,j}, a), \quad (5)$$

including canonical value and aliases, neural payload, typed routing key, uncertainty, and proposed operation or tombstone intent. A trusted assembler then constructs the complete candidate row,

$$\hat{r}_{n,j,a} = \text{AssembleTrusted}(\hat{c}_{n,j,a}, \text{tenant}, \text{ACL}, \text{old protection}, \text{next version}, \text{provenance}, \text{rollback lineage}). \quad (6)$$

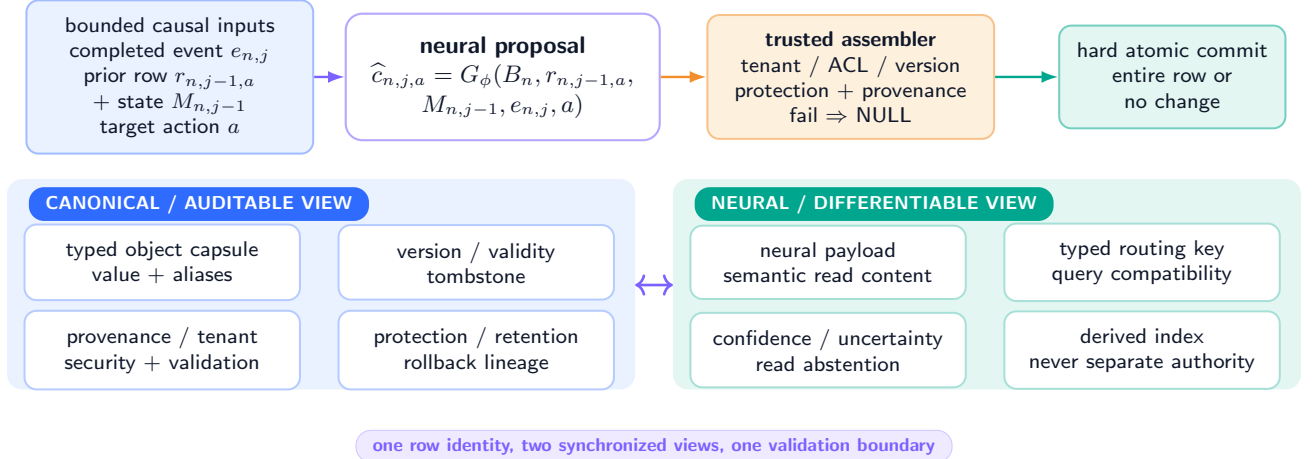
Tenant, ACL/security scope, monotone version, protection inheritance, source receipt, rollback lineage, and deletion verification are system-owned fields rather than free neural outputs. Schema, security, and version checks run before action selection; an invalid candidate fails to NULL. Aliases live in the capsule or a deterministic derived index, never in a separately mutable authoritative map.

We recommend a dual-view vNext contract: canonical content makes the record auditable, while a synchronized neural payload supports differentiable readout. Three negative interface studies motivate this choice; they do not validate it.

Recommended vNext interface: a complete dual-view authoritative row

PROPOSED / UNVERIFIED

motivated by three semantic-interface failures; not yet qualified



Why this vNext: PM-S1, PM-I1, and PM-I2 preserved physical identity but did not expose compositional semantic content. The canonical view makes truth auditable; the neural view remains trainable.

Figure 4: **Proposed complete-row contract.** Both views share one identity and one validation boundary. This is the recommended successor interface, explicitly unverified.

2.4 One hard authoritative overwrite per event

For each event j , the controller chooses $a_{n,j} \in \{\emptyset, 1, \dots, S\}$ and a hard gate $g_{n,j} \in \{0, 1\}$, with $g_{n,j} = 0$ exactly when $a_{n,j} = \emptyset$. Authoritative state changes discretely:

$$M_{n,j} = \begin{cases} M_{n,j-1}, & g_{n,j} = 0 \text{ (NULL)}, \\ \text{Replace}(M_{n,j-1}, a_{n,j}, \hat{r}_{n,j,a_{n,j}}), & g_{n,j} = 1, \end{cases} \quad M_n = M_{n,K_n}. \quad (7)$$

Insert, update, consolidating update, eviction, and deletion are semantic interpretations of replacing one complete row; a tombstone is itself a versioned row. Merging two authoritative resident rows requires two ordered events or a future bounded transaction, because changing one target cannot silently invalidate a second row. NULL leaves every field unchanged. Protection removes forbidden targets. A soft mixture, if useful, belongs only to a non-authoritative fast neural state and never to the versioned fact table.

Hard event-level transition: one complete row or NULL

the versioned fact table never uses a soft mixture; soft state, if used, is explicitly non-authoritative

canonical capsule + neural payload + typed key + version/validity + protection/retention + provenance/security + tombstone commit together

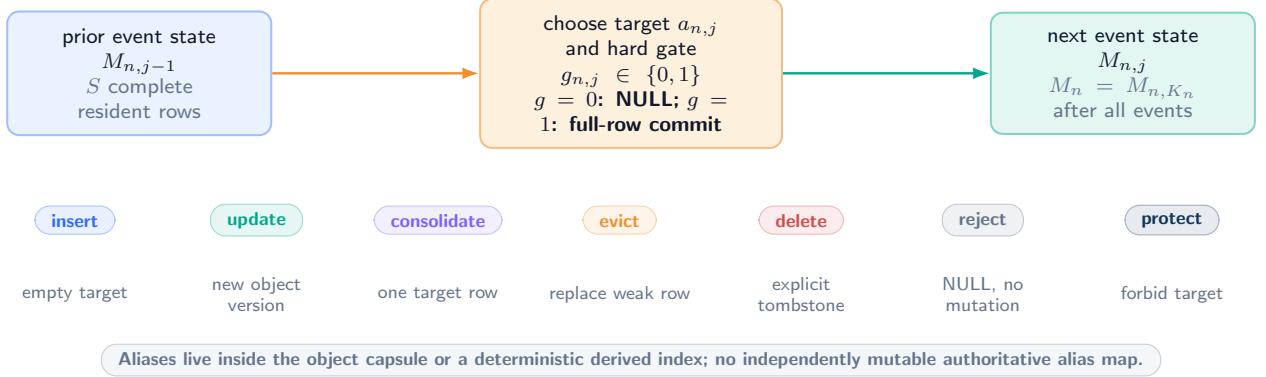


Figure 5: **One physical transition, several lifecycle meanings.** The design does not require a separate mutation primitive for every semantic operation. What matters is the selected target, the replacement row, and the option to do nothing.

If neural payload norms are bounded by B , then a committed payload change satisfies

$$\|v(\hat{r}_{n,j,a_{n,j}}) - v(r_{n,j-1,a_{n,j}})\|_2 \leq 2B. \quad (8)$$

This is a one-step neural-payload bound, not a convergence theorem and not a soft authoritative update. A poor policy can still oscillate or erase useful state.

Worked state transition: Alice moves from Paris to Berlin

ILLUSTRATIVE / NOT MEASURED numbers show the contract, not experimental evidence

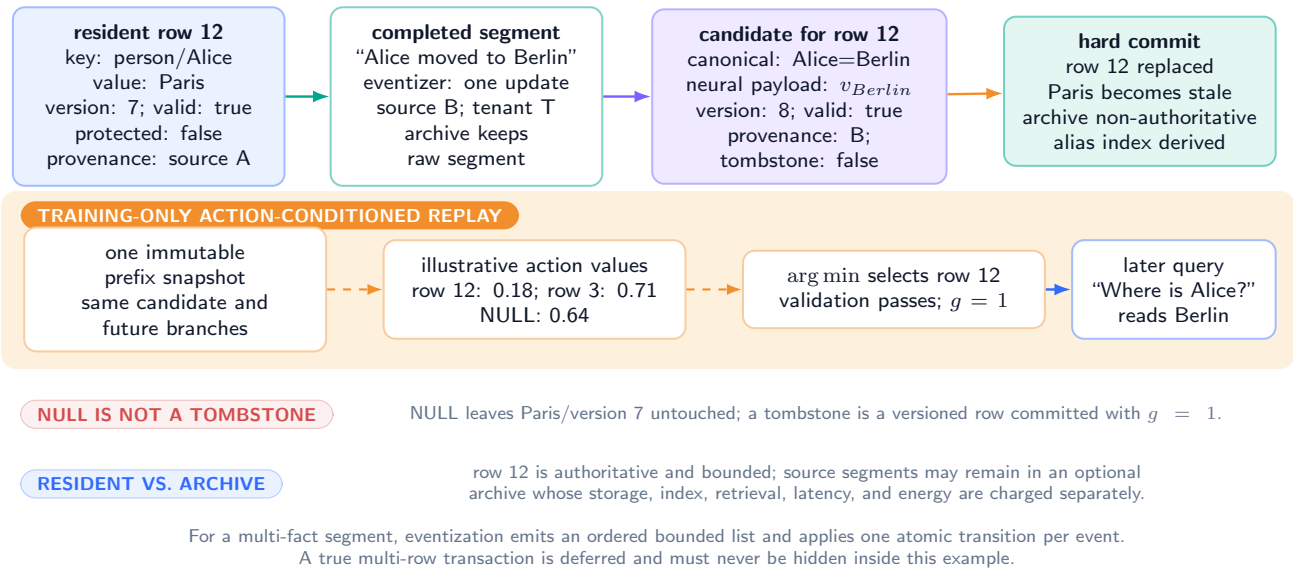


Figure 6: **One complete worked transition.** Alice’s authoritative row advances from Paris/version 7 to Berlin/version 8. The action values are illustrative, not measured. NULL preserves version 7; a tombstone would be a committed version 8 row. The archive remains non-authoritative and separately charged.

2.5 The future prices actions only during training

Let $\sigma_{n,j} = S_\theta(h_{b_n-w+1:b_n}, \bar{B}_n, M_{n,j-1}, e_{n,j})$ be the bounded deployment summary for event j and F a random future. The notation $x_{\leq b_n}$ describes the causal information set only; a deployed controller does not reread the whole prefix unless archive retrieval and its full cost are explicit. Every feasible action constructs its own complete candidate and branches from the same

immutable event-level snapshot. For horizon H ,

$$J_H(a; \sigma_{n,j}, F) = \sum_{k=1}^H \gamma^{k-1} \ell(F_k; M_{n,j}^{(a)}) + \lambda_{\text{int}} D_{\text{collateral}}(a) + \lambda_{\text{write}} \mathbf{1}[a \neq \emptyset]. \quad (9)$$

The deployment target is the conditional value

$$\bar{J}_H(a | \sigma_{n,j}) = \mathbb{E}_{F \sim P(\cdot | \sigma_{n,j})} [J_H(a; \sigma_{n,j}, F)]. \quad (10)$$

Training approximates this expectation with physically grouped continuations that share the same observed prefix. A future-blind value model regresses the detached value for every action,

$$\mathcal{L}_Q = \sum_{a \in \mathcal{A}_{n,j}} \text{Huber}(\hat{Q}_\psi(a | \sigma_{n,j}) - \text{stopgrad } \bar{J}_H(a | \sigma_{n,j})), \quad (11)$$

then chooses $\arg \min_a \hat{Q}_\psi(a | \sigma_{n,j})$ at deployment. Future tokens, answers, and gold actions are absent from the forward API. The first causal intervention may deliberately share one candidate payload and $g_{n,j} = 1$ across slot actions to isolate action value; that controlled simplification is not the complete target-conditioned architecture. A learned latent rollout may later amortize branch construction, but it is neither required for the first identifiable test nor an achieved result.

The information-theoretic intuition is

$$\max_M I(M; X_{\text{future}}) - \beta I(M; X_{\text{past}}), \quad (12)$$

which asks a finite memory to retain predictive information rather than copy the entire past. This is motivation, not an implementation result. Landauer-style reasoning likewise says only that irreversible information processing has physical cost [17]; it supplies no measured hardware-energy advantage for MMLA.

2.6 Compute boundary

With N tokens, local window W , hidden width d , S persistent rows, $K \ll S$ actively read rows, segment length C , routing cost $R_q(S, d)$, and consolidation cost $C_{\text{cons}}(W, S, d)$, an honest context-mixing budget is

$$O(NWd) + O(NR_q(S, d)) + O(NKd) + O\left(\frac{N}{C} C_{\text{cons}}(W, S, d)\right). \quad (13)$$

Eventization, candidate assembly, action replay during training, archive/index work, kernel launches, and memory traffic remain explicit system costs. The proposal predicts a controllable deployment cost only when these interfaces and the consolidation rate stay bounded. Hardware advantage must be measured rather than inferred from asymptotic notation.

3 What the Evidence Does and Does Not Establish

Claim	Accepted basis	Boundary	Status
Causal boundary	Equations 2–4; no deploy future input	Mathematical/interface contract only	FORMALIZED
Controlled lifecycle	Version, lock, tombstone, eviction implementation	Mechanical ancestor	IMPLEMENTED
Lifecycle + typed binding	300/300 and 240/240 per seed under controls	Controlled grammars	VALIDATED
Resident semantic read	PM-S1, PM-I1, PM-I2; PM-I2 0/9 qualified	Frozen interfaces/budgets only	FAILED
Dual-view row	Contract in Figure 4	Negative-evidence motivated	PLANNED
Expected-risk overwrite	Semantic prerequisite failed; oracle not opened	No learned policy result	CLOSED BY GATE
End-to-end cost advantage	No repeated-write or quality–cost crossover	Asymptotic argument only	PLANNED

Table 1: Claim–evidence matrix with one status vocabulary. A positive component does not promote the full architecture.

The first two studies use frozen language-model readers and train only writer, reranking, and calibration components. The relational experiment is a compact jointly trained synthetic compiler. None instantiates the complete architecture in Figure 3. The latest interface study is closer to the frozen native state, but its negative result does not isolate whether optimization, representation, routing, or compositional output geometry is responsible. This separation is deliberate: validated component experiments, an implemented ancestor, the formal architecture, and planned gate-closed successors are four different evidence layers.

4 Controlled Lifecycle Execution

The lifecycle benchmark contains 1,200/300/300 train/dev/test records across six balanced scenarios: same-key overwrite, alias query, stale re-mention, invalid write, protected slot, and capacity pressure. The held-out split is physically disjoint in sentence frames while sharing a disclosed cue vocabulary. A deliberately strong last-mention rule is wrong by construction in four scenarios.

Method	Held-out exact match	Interpretation
Strongest lexical rule	0.333	non-learned baseline
Naive all-writes lifecycle	0.333	no event typing
Learned writer, lifecycle disabled	0.490–0.493	version arbitration removed
Lifecycle active, fallback disabled	0.990	capacity misses remain
Complete controlled path	1.000 on all seeds	full registered system

Table 2: Controlled-language results on 300 held-out records at each of three fixed seeds. Peak resident occupancy is 32 rows. The complete path has zero lifecycle errors, protected-value accuracy 1.0, and recovers all three target evictions.

Lifecycle failures are scenario-specific by construction

exact match · mean of three seeds

	base overwrite	alias query	stale re-mention	invalid write	protected slot	slot pressure
Lexical last-key	1.00	0.00	0.00	0.00	0.00	1.00
Naive lifecycle	1.00	0.00	0.00	0.00	0.00	1.00
Lifecycle disabled	.37	.44	.39	.41	1.00	.33
Fallback disabled	1.00	1.00	1.00	1.00	1.00	.94
Complete path	1.00	1.00	1.00	1.00	1.00	1.00

retrieval baselines succeed only where last mention is valid full path: correct in all six regimes

Figure 7: **Lifecycle performance by scenario.** Lexical and untyped rules work only when the most recent mention is valid. The complete controlled path is exact in all six regimes; removing fallback loses only capacity-pressure cases.

Given selected raw lines, the complete path also answers 300/300 generation records with 146 average prompt tokens. The same frozen Llama-3.1-8B reader given the entire log answers 172/300 with 729 tokens [9]. This comparison demonstrates the value of explicit lifecycle execution in that grammar, not open-language memory management.

5 Selection, Fallback, and the Unknown Future

The multi-hop study uses 1,200 training, 300 calibration, and 300 held-out HotpotQA questions after duplicate filtering [37]. Natural records have median length 933 words. An extended condition adds answer-excluded, train-derived distractors to a median of 8,221 words. The intended evidence targets are 25% and 10% of context words. Frozen Llama-3.1-8B and Qwen2.5-14B serve as readers [9, 26].

Protocol note. The original Llama execution failed its frozen whole-prompt budget gate. Under the intended evidence-word reading, the extended condition satisfies the 10% cap; the natural condition averages 22.57% but reaches 61.36% on one short record because of the original first-passage exception. The Qwen follow-up uses the corrected strict per-record packer and is the budget-qualified execution. Llama quality rows are retained as failed-protocol evidence rather than a second successful budget qualification.

Reader	Method	Natural F1	tokens	Extended F1	tokens
Llama-3.1-8B	Full context	.714	1,477	.666	12,024
	Dense, budget matched	.516	432	.622	1,300
	cache + archive fallback	.659–.663	434	.677–.684	1,323
Qwen2.5-14B	Full context	.805	1,521	.715	12,646
	Dense, budget matched	.594	425	.753	1,338
	cache + archive fallback	.755–.760	428	.813–.830	1,367

Table 3: Held-out answer quality and average whole-prompt tokens. The learned path improves over the registered dense baseline by 14.3–16.6 F1 at natural length and 5.5–7.6 F1 at extended length. The archive storage, indexing, retrieval latency, and energy are excluded.

Held-out QA: answer quality at a fraction of full-context active tokens

held-out questions · average whole-prompt tokens

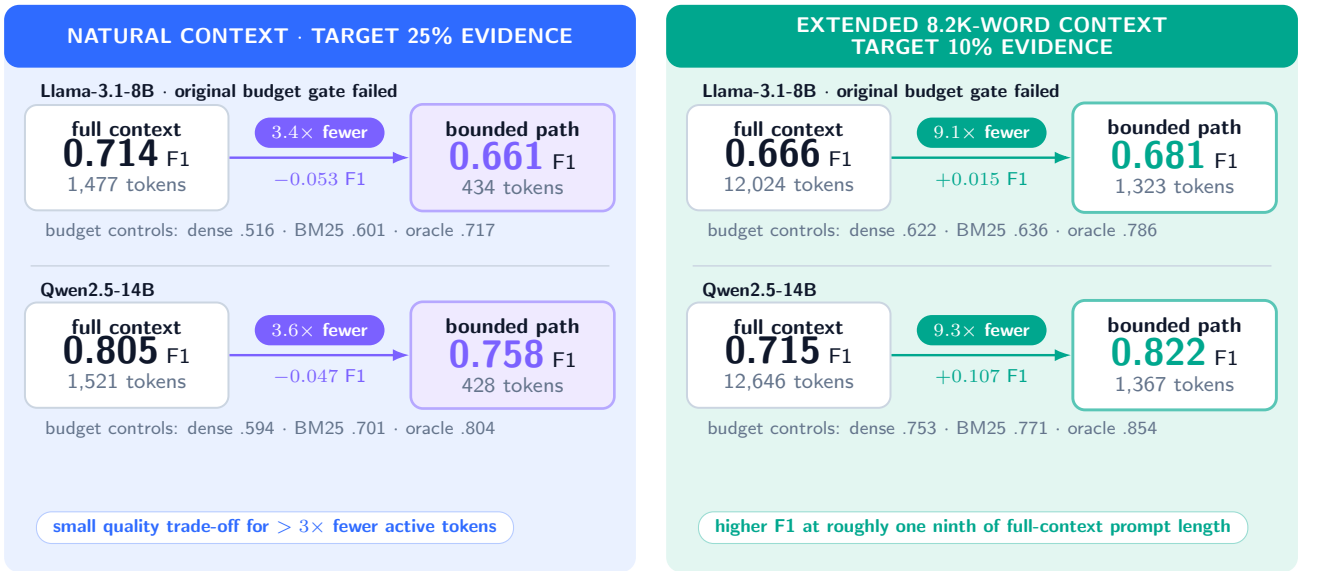


Figure 8: **Held-out QA: answer quality versus active prompt tokens.** At natural length, selection trades a small amount of F1 for about 3.4–3.6× fewer active tokens. At 8.2k words it improves F1 while using roughly one ninth as many prompt tokens. The Llama rows are retained failed-protocol evidence; the Qwen rows use the corrected strict packer. Archive and retrieval costs are excluded.

The learned selector creates the quality margin; the 32-slot resident cache mainly preserves it. Fallback searches the retained full context on 39% of natural and 72% of extended records. This is evidence for bounded hot state plus abstention, not bounded total storage. The margin over BM25 is a more demanding 4.0–6.2 F1. A trained retriever and end-to-end archive cost remain missing. The deeper lesson is an information boundary: passage-only salience is only modestly predictive of which passage a future question will need, so a writer must be allowed to abstain.

6 Typed Relational Binding

The controlled lifecycle parser assumes a structured event representation. AFRT asks whether a differentiable compiler can recover role–filler binding when content and absolute position do not compose. It predicts semantic anchors and transports their posterior mass through typed relation paths before dynamic partial Sinkhorn assignment [29, 7, 33, 28, 2]:

$$p(f_r = c \mid x, e) = \sum_a p(a_r = a \mid x, e) T_\theta(c \mid a, r, x, e). \quad (14)$$

Real filler capacity is at most one, each role has its own NULL, and relation direction and path steps share parameters.

AFRT: factor semantic anchoring from relation-conditioned filler transport

all four decoders share one encoder, one training trajectory, and one checkpoint

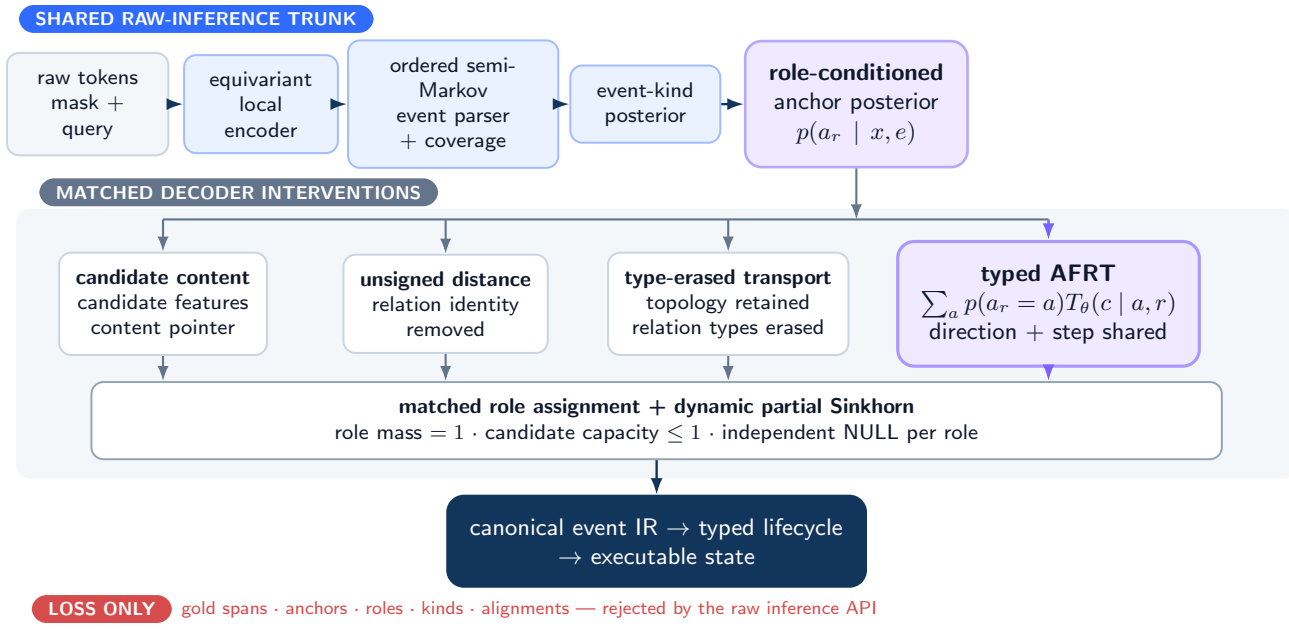


Figure 9: **AFRT and matched decoders.** Four decoders share one encoder, training trajectory, and checkpoint. Gold spans, anchors, roles, kinds, and alignments are loss-only and absent from raw inference.

Each of three fresh data/model seeds has 512 fitting and 240 physically disjoint held-out records, path lengths 2–8, distractor bridges, 2–6 events, and optional or NULL roles. The typed decoder reaches 240/240 held-out whole-record exactness and 40/40 on all six stress strata at every seed. Candidate-content, unsigned-distance, and type-erased transport decoders from the same checkpoints each obtain 0/240 whole-record successes. The paired gain is therefore +240 against every registered control at every seed.

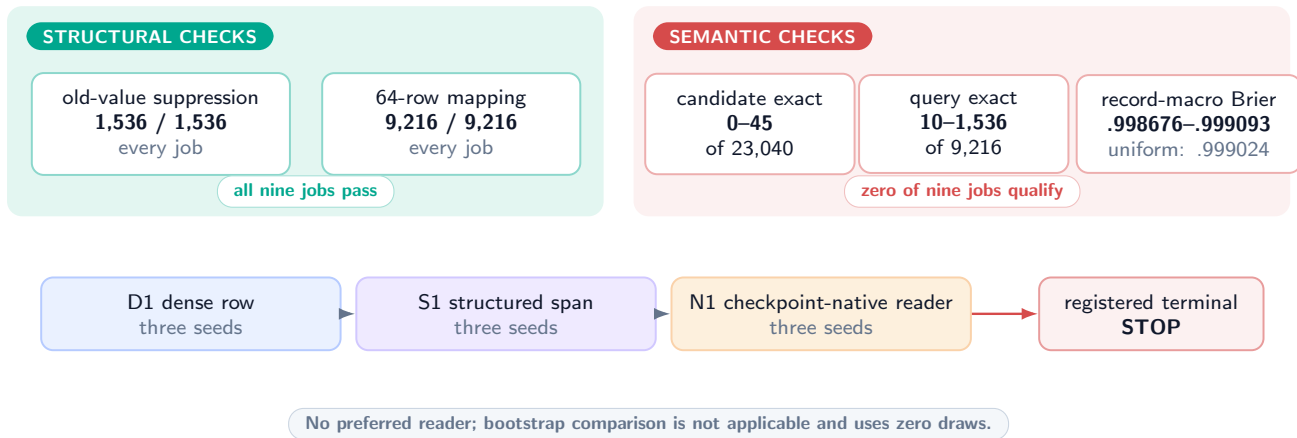
This is matched structural evidence: typed topology supplies a binding signal absent from those controls in the registered synthetic setting. Co-adaptation through the shared encoder is possible, and the experiment lacks a type-aware non-factorized control. It does not establish natural-language relation discovery, arbitrary graph reasoning, long-context scaling, or a full-model advantage.

7 The Semantic Interface Failure

Three preregistered studies attempted to read semantic content from V28, the frozen 352.3M-parameter model-only checkpoint produced by the 50.0M-token native-scaffold pilot. Its 1,025 output classes comprise ordered 32×32 atom pairs (classes 0–1023) plus one ABSENT class (1024). The latest, PM-I2, directly compares a dense row (D1), a structured 4×192 span (S1), and a checkpoint-native reader (N1). Each arm uses three fixed seeds, 2,048 updates per job, final-only checkpoints, and a fixed Latin order. Predictions were materialized before evaluator truth. Nine checkpoint bundles, nine factor bundles, nine metric bundles, and 55,296 raw rows were independently audited.

PM-I2: structure passes; semantics remain near chance

one frozen V28 release · three readers · three seeds · 55,296 raw rows



This rejects the three frozen interfaces under their matched budgets. It does not prove that V28 contains no useful latent memory information.

Figure 10: **PM-I2 registered negative result.** Physical mapping and old-value suppression pass, yet semantic prediction remains approximately uniform/chance. Zero jobs qualify, no arm preference is defined, and the registered bootstrap comparison is not applicable.

Reader family	Jobs	Candidate exact	Query exact	Brier range	Qualified
D1 dense row	3	0/23,040 each	1,536/9,216 each	.998815–.998981	0/3
S1 structured span	3	0–21/23,040	10–1,536/9,216	.998676–.999035	0/3
N1 native reader	3	30–45/23,040	10–645/9,216	.998992–.999093	0/3

Table 4: PM-I2 summary. The 1,025-class uniform Brier reference is $1024/1025 = 0.999024$. Every job passes 1,536/1,536 old-value suppression and 9,216/9,216 row-mapping consistency. Those structural checks are not semantic success.

A separately implemented within-project adjudicator reproduced all nine false decisions and published the registered “no current interface qualified” terminal. Because fewer than two arms qualify, bootstrap comparison is not applicable and uses zero draws. The result rejects these three interfaces under their frozen budgets. It does not prove that V28 contains no useful latent memory information and does not tell us whether the dominant failure lies in fitting, routing, representation, or output composition. The exact machine identifier is retained in the provenance metadata.

The next scientific step is therefore diagnostic, not scale. PM-I3 is a fault-localization tree, not a reader-selection contest. It reuses immutable PM-I2 finals without training updates and asks, in order: (i) does a reader fit its registered examples; (ii) does it compose on fresh pairs; (iii) does a routing oracle repair output; (iv) does a content oracle repair output; or (v) does the full qualification pass? At this manuscript’s evidence cutoff, PM-I3 has no admitted learned result and its interrupted transport recovery is not evidence. Predictive overwrite, larger pretraining, and reader selection remain closed.

The next interface comparison should prioritize canonical token span, typed canonical tuple, dual-view row, and pure native latent row under the same checkpoint, data, exposure, active parameters, seeds, and final-only evaluation. Only if every semantic-interface gate passes may the expected-risk oracle be opened. The oracle itself must show a preregistered margin before Q_ψ is trained. A future-blind controller must then report action regret, top-1 agreement, NULL precision/recall, uncertainty, stale leakage, and collateral damage. Repeated-write trajectories follow only after that; a GPU/system quality–cost crossover follows later still.

8 Relation to Existing Memory Systems

Differentiable addressable memory begins with Memory Networks and Neural Turing Machines [35, 10]. Transformer-XL, Compressive Transformers, Memorizing Transformers, LongMem, retrieved-text models, and learned memory layers preserve or recover information across longer horizons in different forms [8, 27, 36, 34, 18, 5, 3]. KV-retention methods and landmark mechanisms choose which cached representations remain accessible [41, 19, 22].

MMLA focuses on a narrower missing operation: deciding whether a completed observation should change a bounded authoritative state. MemGPT treats memory as a systems hierarchy and virtual context [23]. MemOps provides diagnostic memory operations rather than prescribing an architecture [15]. Metis is a closer architecture-level comparator, with persistent native state, memory-specific training, and forward state updates; its reported trajectory degradation and interference reinforce the need for explicit overwrite and abstention [40]. FlashKDA is a high-performance recurrent-state kernel substrate rather

than a semantic memory policy; it may be complementary if an integrated benchmark eventually supports that claim [6]. We claim neither superiority nor an implemented integration with these systems.

World models and latent imagination motivate compressed prediction and control [13, 14]; complementary learning systems motivate fast episodic state and slower parameter integration [21]. Our use is more specific: grouped actual futures supervise the value of one bounded memory action, while deployment remains causal.

9 A Conditional Roadmap, Not a Promise

MMLA roadmap: localize first, open each gate once

all boxes below the registered negative are FUTURE / UNVERIFIED; arrows are stop rules, not promises

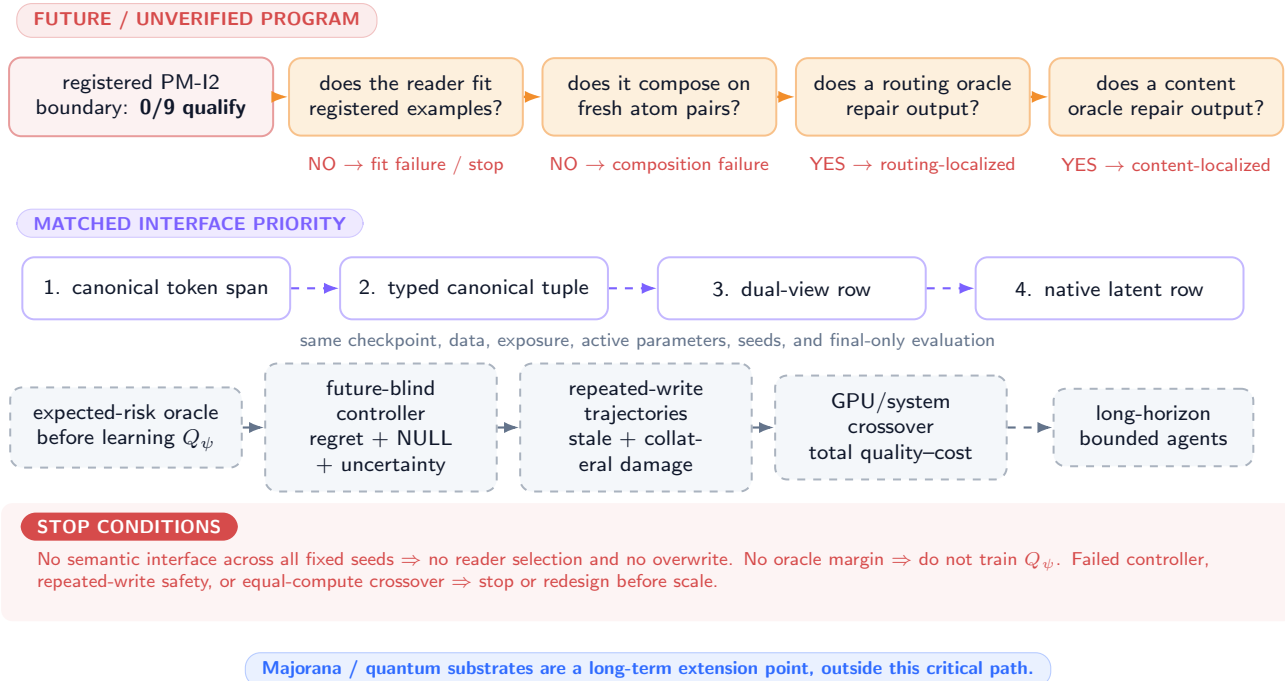


Figure 11: **Research roadmap.** Dashed boxes are proposed or future work. Advancement is conditional on the preceding frozen matched test. The graphic is a vision of what a successful program could unlock, not a diagram of a completed model.

The roadmap has explicit stop conditions:

1. **Diagnose before redesign.** A fit failure stops generalization claims. A fresh-composition failure stops reader selection. Routing- and content-oracle repairs localize the defect but do not qualify a reader.
2. **Qualify a semantic interface.** Reconstruction, query, calibration, preservation, stale suppression, and mapping must all pass on fresh data across fixed seeds. If no interface qualifies, overwrite remains closed.
3. **Test the expected-risk oracle first.** Replay every feasible action from one snapshot. No registered oracle margin means no training of Q_ψ .
4. **Train the future-blind controller only after the oracle.** Report regret, top-1 action agreement, NULL precision/recall, uncertainty, stale leakage, and collateral damage. Failure stops or redesigns the policy.
5. **Evaluate repeated writes before systems scale.** Test churn, rollback, protected rows, adversarial writes, and long-horizon state trajectories before any quality-cost claim.
6. **Scale only after an equal-compute crossover.** Include archive storage/index/retrieval, replay, kernels, memory traffic, latency, and energy. Majorana or quantum backends remain long-term extension points outside this critical path.

This program could yield a useful middle time scale between context and weights. It may become more valuable as the backbone becomes better at predicting future use. That is an empirical scaling hypothesis, not an AGI theorem. General intelligence also requires planning, exploration, causal models, calibration, safety, social reasoning, and stable parameter-level learning.

An invitation to reproduce and challenge the bottleneck

The most useful community contribution is not another unregistered scale run. It is an independent reproduction or falsification of the semantic-interface boundary: Can a frozen native memory row be decoded on fitted examples? Does failure emerge only under new atom composition? Does a routing oracle help? Does a content oracle help? Can a reader preserve unrelated rows while suppressing a stale value? We encourage alternative dense, structured, span-native, and checkpoint-native interfaces under equal data, updates, seeds, checkpoint, and final-only evaluation. Negative replications are as valuable as positive ones if the contract is fixed before evaluator truth is opened.

10 System Contract, Limitations, and Safety

The positive evidence is controlled. Lifecycle uses a grammar with shared cue words. QA uses static passages and a retained archive; it does not test overwrite. AFRT uses a synthetic compiler and permits shared-encoder co-adaptation. None demonstrates a fully pretrained memory-native language model. PM-I2 provides a strong negative boundary, but it does not localize the cause.

The architecture also creates risks. A persistent writer can be poisoned, flooded, or tricked into deleting protected information. Cross-user state can leak; aliases can collide; rollback can resurrect stale values. These concerns belong inside the system contract, not only in a limitations list. Every complete row therefore carries provenance, tenant/security scope, validity, protection, retention, tombstone, validation status, and rollback lineage. The action mask enforces protection before value comparison; a failed schema, security, or monotone-version check returns NULL. Recovery restores a prior versioned snapshot and invalidates any deterministic derived index. Verified deletion writes a tombstone and applies the declared archive policy; it is not silently conflated with NULL. These mechanisms are formal requirements and remain unvalidated end to end.

Training-time branch replay may be expensive even if deployment is cheap. The grouped futures must represent the intended deployment distribution. A controller can overfit synthetic branch structure, and a lower language-model loss can hide collateral memory damage. The correct comparison must therefore include old-value leakage, unrelated-fact preservation, action regret, memory churn, latency, energy, and archive cost.

11 Conclusion

Memory management asks which observations should be allowed to change persistent state. MMLA proposes a causal answer: reinterpret only completed local segments, use real futures only as training supervision, and commit one complete-row overwrite or NULL. The mathematics gives a coherent and testable object, not proof that it works.

The evidence is intentionally mixed. Controlled lifecycle, selection with fallback, and typed relational transport each succeed within registered boundaries. The matched current-state interface study fails across all nine jobs despite stable physical mapping. The complete predictive loop therefore remains unvalidated and closed behind two gates: a semantic same-checkpoint interface and a future-valued overwrite policy that survives equal-compute testing. Publishing both the positive components and the negative boundary makes the next question sharper. We invite the community to reproduce, criticize, and improve that interface before more scale is spent.

The architecture is formalized, but its same-checkpoint semantic memory interface and learned predictive overwrite policy are not yet validated.

AI-Assisted Tooling Disclosure

The authors used AI-assisted programming and writing tools for code development, experiment execution, drafting, and editing. The authors reviewed and verified the experimental outputs, claims, and manuscript content and take responsibility for the work.

Author Contributions

Junyi Zou and Avrova Donz contributed equally to this work. Both authors share responsibility for the research and manuscript.

References

- [1] Alexander A. Alemi, Ian Fischer, Joshua V. Dillon, and Kevin Murphy. Deep variational information bottleneck. <https://arxiv.org/abs/1612.00410>, 2016.

- [2] Peter W. Battaglia, Jessica B. Hamrick, Victor Bapst, Alvaro Sanchez-Gonzalez, Vinicius Zambaldi, Mateusz Malinowski, Andrea Tacchetti, David Raposo, Adam Santoro, Ryan Faulkner, Caglar Gulcehre, Francis Song, Andrew Ballard, Justin Gilmer, George Dahl, Ashish Vaswani, Kelsey Allen, Charles Nash, Victoria Langston, Chris Dyer, Nicolas Heess, Daan Wierstra, Pushmeet Kohli, Matthew Botvinick, Oriol Vinyals, Yujia Li, and Razvan Pascanu. Relational inductive biases, deep learning, and graph networks. <https://arxiv.org/abs/1806.01261>, 2018.
- [3] Vincent-Pierre Berges et al. Memory Layers at Scale. <https://arxiv.org/abs/2412.09764>, 2024. arXiv:2412.09764.
- [4] William Bialek, Ilya Nemenman, and Naftali Tishby. Predictability, complexity, and learning. *Neural Computation*, 13(11):2409–2463, 2001.
- [5] Sebastian Borgeaud et al. Improving Language Models by Retrieving from Trillions of Tokens. <https://arxiv.org/abs/2112.04426>, 2021. arXiv:2112.04426.
- [6] Yutian Chen, Zhiyuan Li, Yucheng Wang, and Ming Wei. FlashKDA: Flash kimi delta attention. <https://github.com/MoonshotAI/FlashKDA>, 2026. Source repository and benchmark documentation, accessed 11 August 2026.
- [7] Marco Cuturi. Sinkhorn distances: Lightspeed computation of optimal transport. In *Advances in Neural Information Processing Systems*, volume 26, 2013.
- [8] Zihang Dai et al. Transformer-XL: Attentive Language Models Beyond a Fixed-Length Context. <https://arxiv.org/abs/1901.02860>, 2019. arXiv:1901.02860.
- [9] Aaron Grattafiori et al. The Llama 3 Herd of Models. <https://arxiv.org/abs/2407.21783>, 2024.
- [10] Alex Graves, Greg Wayne, and Ivo Danihelka. Neural turing machines. <https://arxiv.org/abs/1410.5401>, 2014.
- [11] Albert Gu and Tri Dao. Mamba: Linear-Time Sequence Modeling with Selective State Spaces. <https://arxiv.org/abs/2312.00752>, 2023. arXiv:2312.00752.
- [12] Albert Gu, Karan Goel, and Christopher Ré. Efficiently modeling long sequences with structured state spaces. In *International Conference on Learning Representations*, 2022.
- [13] David Ha and Jürgen Schmidhuber. World models. <https://arxiv.org/abs/1803.10122>, 2018.
- [14] Danijar Hafner, Timothy Lillicrap, Jimmy Ba, and Mohammad Norouzi. Dream to control: Learning behaviors by latent imagination. In *International Conference on Learning Representations*, 2020.
- [15] Xixuan Hao, Zeyu Zhang, Zehao Lin, Yihang Sun, Ziliang Guo, Xichong Zhang, Yuxuan Liang, Feiyu Xiong, and Zhiyu Li. MemOps: Benchmarking lifecycle memory operations in long-horizon conversations. <https://arxiv.org/abs/2607.12893>, 2026.
- [16] Angelos Katharopoulos et al. Transformers are RNNs: Fast Autoregressive Transformers with Linear Attention. <https://arxiv.org/abs/2006.16236>, 2020. arXiv:2006.16236.
- [17] Rolf Landauer. Irreversibility and heat generation in the computing process. *IBM Journal of Research and Development*, 5(3):183–191, 1961.
- [18] Patrick Lewis et al. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. <https://arxiv.org/abs/2005.11401>, 2020. arXiv:2005.11401.
- [19] Yuhong Li, Yingbing Huang, Bowen Yang, Bharat Venkitesh, Acyr Locatelli, Hanchen Ye, Tianle Cai, Patrick Lewis, and Deming Chen. SnapKV: LLM knows what you are looking for before generation. <https://arxiv.org/abs/2404.14469>, 2024.
- [20] Michael L. Littman, Richard S. Sutton, and Satinder Singh. Predictive representations of state. In *Advances in Neural Information Processing Systems*, volume 14, pages 1555–1561, 2001.
- [21] James L. McClelland, Bruce L. McNaughton, and Randall C. O’Reilly. Why there are complementary learning systems in the hippocampus and neocortex: Insights from the successes and failures of connectionist models of learning and memory. *Psychological Review*, 102(3):419–457, 1995.
- [22] Amirkeivan Mohtashami and Martin Jaggi. Landmark attention: Random-access infinite context length for transformers. <https://arxiv.org/abs/2305.16300>, 2023.
- [23] Charles Packer, Sarah Wooders, Kevin Lin, Vivian Fang, Shishir G. Patil, Ion Stoica, and Joseph E. Gonzalez. MemGPT: Towards LLMs as operating systems. <https://arxiv.org/abs/2310.08560>, 2023.
- [24] Bo Peng, Eric Alcaide, Quentin Anthony, Alon Albalak, Samuel Arcadinho, Stella Biderman, Huanqi Cao, Xin Cheng, Michael Chung, Matteo Grella, Kranthi Kiran Gv, Xuzheng He, Haowen Hou, Przemyslaw Kazienko, Jan Kocon, Jiaming Kong, Bartłomiej Koptyra, Hayden Lau, Krishna Sri Ipsit Mantri, Ferdinand Mom, Atsushi Saito, Xiangru Song, Haowen Tang, Bolun Wang, Johan S. Wind, Stanislaw Wozniak, Ruichong Zhang, Zhenyuan Zhang, Qihang Zhao, and Peng Zhou. RWKV: Reinventing RNNs for the transformer era. <https://arxiv.org/abs/2305.13048>, 2023.
- [25] Michael Poli, Stefano Massaroli, Eric Nguyen, Daniel Y. Fu, Tri Dao, Stephen Baccus, Yoshua Bengio, Stefano Ermon, and Christopher Ré. Hyena hierarchy: Towards larger convolutional language models. <https://arxiv.org/abs/2302.10866>, 2023.
- [26] Qwen et al. Qwen2.5 Technical Report. <https://arxiv.org/abs/2412.15115>, 2024.
- [27] Jack W. Rae et al. Compressive Transformers for Long-Range Sequence Modelling. <https://arxiv.org/abs/1911.05507>, 2019. arXiv:1911.05507.
- [28] Adam Santoro, David Raposo, David G. T. Barrett, Mateusz Malinowski, Razvan Pascanu, Peter Battaglia, and Timothy Lillicrap. A simple neural network module for relational reasoning. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

- [29] Sunita Sarawagi and William W. Cohen. Semi-markov conditional random fields for information extraction. In *Advances in Neural Information Processing Systems*, volume 17, 2004.
- [30] Yutao Sun et al. Retentive Network: A Successor to Transformer for Large Language Models. <https://arxiv.org/abs/2307.08621>, 2023. arXiv:2307.08621.
- [31] Naftali Tishby, Fernando C. Pereira, and William Bialek. The information bottleneck method. <https://arxiv.org/abs/physics/0004057>, 2000.
- [32] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30, 2017.
- [33] Oriol Vinyals, Meire Fortunato, and Navdeep Jaitly. Pointer networks. In *Advances in Neural Information Processing Systems*, volume 28, 2015.
- [34] Weizhi Wang et al. Augmenting Language Models with Long-Term Memory. <https://arxiv.org/abs/2306.07174>, 2023. arXiv:2306.07174.
- [35] Jason Weston, Sumit Chopra, and Antoine Bordes. Memory networks. <https://arxiv.org/abs/1410.3916>, 2014.
- [36] Yuhuai Wu, Markus N. Rabe, DeLesley Hutchins, and Christian Szegedy. Memorizing Transformers. <https://arxiv.org/abs/2203.08913>, 2022. arXiv:2203.08913.
- [37] Zhilin Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William W. Cohen, Ruslan Salakhutdinov, and Christopher D. Manning. HotpotQA: A dataset for diverse, explainable multi-hop question answering. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 2369–2380, Brussels, Belgium, 2018. Association for Computational Linguistics. <https://aclanthology.org/D18-1259/>.
- [38] Jingyang Yuan, Huazuo Gao, Damai Dai, Junyu Luo, Liang Zhao, Zhengyan Zhang, Zhenda Xie, Y. X. Wei, Lean Wang, Zhiping Xiao, Yuqing Wang, Chong Ruan, Ming Zhang, Wenfeng Liang, and Wangding Zeng. Native sparse attention: Hardware-aligned and natively trainable sparse attention. <https://arxiv.org/abs/2502.11089>, 2025.
- [39] Manzil Zaheer et al. Big Bird: Transformers for Longer Sequences. In *Advances in Neural Information Processing Systems*, volume 33, 2020. <https://arxiv.org/abs/2007.14062>.
- [40] Zeyu Zhang, Ziliang Guo, Yihang Sun, Xichong Zhang, Xixuan Hao, Zehao Lin, Yang Zhang, Xiaoyan Zhao, Tong Shen, Bo Tang, Zhi-Qin John Xu, Junchi Yan, Haofen Wang, Xu Chen, Feiyu Xiong, Zhiyu Li, and Tat-Seng Chua. Metis: Memory foundation model. <https://arxiv.org/abs/2607.26760>, 2026.
- [41] Zhenyu Zhang, Ying Sheng, Tianyi Zhou, Tianlong Chen, Lianmin Zheng, Ruisi Cai, Zhao Song, Yuandong Tian, Christopher Ré, Clark Barrett, and Zhangyang Wang. H2O: Heavy-hitter oracle for efficient generative inference of large language models. <https://arxiv.org/abs/2306.14048>, 2023.

A Evidence Provenance and Frozen Boundaries

This v3 replacement is self-contained. The separately reviewed companion Technical Report at <https://github.com/MMLA-org/mmla-memory> preserves the extended derivations, experiment history, failures, and audit detail; no claim in this manuscript depends on the companion document. Table 5 records the frozen project evidence used for the most consequential claim boundaries.

Evidence	Authority or reproducibility record	SHA-256
Evidence admission	frozen project audit record	3b53f993...579
Lifecycle and QA	frozen result rows, summaries, and preregistration records	see retained manifests
AFRT confirmation	<code>paper_evidence_r5p5_20260803</code> result table, terminal audit, claim update, manifest	package checksums
PM-S1 aggregate	external semantic adapter negative	6167308d...995
PM-I1 aggregate	raw representation probe negative	c0ff1fb3...d5d
PM-I2 aggregate	registered no-interface terminal	65ba2236...9ee
PM-I2 bootstrap	not applicable; zero draws	1cb09ce3...cb6f

Table 5: Abbreviated evidence provenance. Full paths, modes, sizes, and hashes are retained in the project audit records. Producer success booleans were not admitted without separate reduction.

The Memory-knot logo, notation, and figure language are preserved in this source package. The public repository is <https://github.com/MMLA-org/mmla-memory>.

B Formal Properties and Non-Claims

Causality. At boundary b_n , z_n is a function only of $(X_{\leq b_n}, M_{n-1})$, and M_n is unavailable to positions $t \leq b_n$. Bidirectionality inside the completed segment therefore introduces no path from unobserved future tokens to an already emitted prediction.

Atomicity. For every event j , Equation 7 gives $\text{rank}(V(M_{n,j}) - V(M_{n,j-1})) \leq 1$ and $\|V(M_{n,j}) - V(M_{n,j-1})\|_{0,\text{row}} \leq 1$, where V stacks neural payload views. Across a segment, $\|V(M_n) - V(M_{n-1})\|_{0,\text{row}} \leq K_n$. Every authoritative field of the selected row commits together.

Greedy value error. Let $a^* = \arg \min_a \bar{J}_H(a | s)$ and $\hat{a} = \arg \min_a \hat{Q}(a | s)$. If $\max_a |\hat{Q}(a | s) - \bar{J}_H(a | s)| \leq \epsilon$, then

$$\bar{J}_H(\hat{a} | s) - \bar{J}_H(a^* | s) \leq 2\epsilon.$$

This motivates held-out value error and action regret. It does not guarantee that finite grouped continuations represent deployment.

Non-claims. None of the equations proves learnability, safety, intelligence, biological plausibility, a hardware speedup, or superiority over a fully trained long-context model. Those remain empirical questions governed by Figure 11.

C Reproduction Questions

Internal hashes prove consistency inside the retained evidence set; they are not external reproducibility. The intended public artifact roadmap is therefore stronger than a checksum list: deterministic data and row constructors; PM-I2 checkpoints and immutable manifests; truth-free prediction factors materialized before evaluator access; an independent evaluator; a pinned environment; one terminal reproduce command; and an artifact dependency graph linking source, data, hidden state, checkpoint, factors, rows, summaries, and decisions. Those assets are not yet represented as publicly downloadable in this manuscript.

The public repository intentionally contains only a README and the separately reviewed complete companion Technical Report. It does not pretend that the experimental assets above are already public. We recommend that independent implementations report at least:

1. exact checkpoint, tokenizer, data, seed, update, and final-only selection bindings;
2. fitted-example reconstruction before fresh-composition evaluation;
3. routing-oracle and content-oracle diagnostics that cannot change the qualification decision;
4. old-value suppression, unrelated-fact preservation, missing-query behavior, calibration, and physical-row invariance;
5. raw per-record outputs materialized before evaluator truth is opened;
6. total cost including archive, index, replay, kernel, and memory-traffic overhead;
7. frozen failure criteria and a commitment not to replace failed seeds or select a preferred reader from unqualified systems.

The open problem is deliberately concrete: build a semantic interface that generalizes beyond its fit pairs while preserving the lifecycle invariants already demonstrated mechanically. A solution would unlock the first matched expected-risk overwrite test; failure would provide a principled reason to redesign the resident representation before joint pretraining. We invite replications, adversarial audits, alternative interfaces, and negative results under a frozen contract.